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Accurate prediction of black rot epidemics in vineyards using a weather-driven disease model

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Abstract

BACKGROUND: Grapevine black rot caused by *Guignardia bidwellii* is a serious threat in vineyards, especially in areas with cool and humid springs. A mechanistic, weather-driven model was recently developed for the detailed prediction of black rot epidemics. The aim of this work was to evaluate the model by comparison with observed disease development in leaves and clusters in a vineyard in north Italy from 2013 to 2015.

RESULTS: The model accurately predicted disease onset. The probability of predicting new infections that did not occur (i.e. unjustified alarms) was ≤0.180, while the probability of missing actual infections was 0.175 for leaves and 0.263 for clusters. In 78% of these false negative predictions, the difference between expected and actual disease onset was ±2 days; therefore, only one infection period was actually missed by the model. The model slightly overestimated disease severity (mainly on leaves) when the observed disease severity was >0.6.

CONCLUSION: The model was highly accurate and robust in predicting the infection periods and dynamics of black rot epidemics. The model can be used for scheduling fungicide sprays in vineyards.
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Keywords: epidemiological model; decision support system; disease management; *Guignardia bidwellii*

1 INTRODUCTION

Grapevine black rot, caused by the ascomycete *Guignardia bidwellii* (Ellis) Viala and Ravaz [anamorph: *Phyllosticta ampelicida* (Englem.) van der Aa], is a serious threat in vineyards, especially in grape-growing areas characterised by cool and moist weather in spring and early summer.[1,2]

Black rot originates from the United States.[3] In Europe, black rot was first detected in France in 1885[2,4–6] and subsequently spread to other grape-growing areas on the continent.[7,8] In Italy, *G. bidwellii* was first detected in Tuscany at the end of the nineteenth century.[9] Reoccurrences of black rot epidemics were recently observed in Friuli (Bigot G, private communication) and Tuscany,[10] where the disease has been considered endemic.

G. bidwellii can infect green tissues of vinegrapes, including leaves, petioles, tendrils and shoots. On these tissues, the disease appears as necrotic light-brown lesions containing small black pycnidia.[11–13] Berries are susceptible from mid-bloom and thereafter.[14] Early symptoms on berries are circular white spots surrounded by a reddish-brown ring; these spots rapidly enlarge until the entire berry rots and is transformed into a shrivelled, hard, blue-black mummy covered with pycnidia and perithecia.[1,10,14–16]

Black rot can reduce yields by up to 80% and can substantially reduce wine quality.[1,4,13,17,18] Under favourable conditions, repeated infection cycles can develop between grapevine bud break and berry ripening; these cycles are caused by ascospores and conidia from the overwintered inoculum sources (berry mummies and cane lesions) and by conidia produced on leaf and shoot lesions and on rotted berries.[1] Black rot is characterised by a long incubation period; up to 3 weeks on young fruit and leaves and up to 5 weeks shortly before berries become resistant.[14,19]

Black rot has been traditionally controlled by repeated application of fungicides on a calendar basis from the early shoot-growth stage through veraison.[20] The most important fungicide applications are (i) when new shoots are 2–3 cm long, (ii) just before bloom and (iii) just after bloom, when the fruit has set.[19] After fruit set, applications continue at about 10 day intervals as long as the weather is rainy and humid; sprays can be discontinued when the weather turns dry.[21] To optimise fungicide timing, researchers have developed simple disease forecasting models based on the length of the wetness period and the temperature conditions required for infection.[17,22–25]

A new, mechanistic, dynamic, weather-driven model was recently developed for the detailed prediction of black rot epidemics.[26] The model accounts for the complexity of the life cycle of *G. bidwellii*. Model output was compared in broad terms with three typical, historical black rot epidemics, indicating that the model reliably describes disease development.[26] However, detailed, specific model validation is still lacking. The present

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work was therefore conducted to determine the accuracy of the model in predicting black rot epidemics.

2 MATERIALS AND METHODS

2.1 Vineyard site and characteristics

A study was carried out in a vineyard on the campus of the University of Piacenza (north Italy, 45° 2′ N latitude, 9° 43′ E longitude) from 2013 to 2015. The vines, which were cv. Barbera, were 7 years old in 2013 and were trained with a Guyot system. A single, 15 m long row was used in this work. No fungicides were applied to any of the vineyard during the experiment.

A standard weather station (Vantage Pro2; Davis Instruments Corp., Hayward, CA) located in the vineyard provided hourly records of air temperature (T, °C), relative humidity (RH, %), rainfall (Rain, mm) and leaf wetness (LW, min).

In late February of each year of the 3 year experiment, *G. bidwellii* inoculum was introduced into the experimental vineyard. The inoculum consisted of mummified black rot berries that had been collected at the end of the previous growing season from three to five commercial vineyards; these commercial vineyards were located in north Italy and were severely affected with black rot. The collected mummies were kept outdoors during the winter. In the following February, the mummies were mixed to form one composite sample before a 20 g subsample was placed in each of nine containers made of metallic screen with 90 mm wide openings. The containers were attached to the trellis and were 1.80 m above the ground and 1.5 m apart along the grapevine row. The mummies were left in position throughout the growing season.

2.2 Disease assessment

Disease was assessed in all the leaves and clusters of the vines from when the first leaf unfolded (stage 11 of the BBCH scale of Lorenz *et al.*[27]) to the softening of berries (stage BBCH 85). This is the period during which plants are susceptible to *G. bidwellii*.[14]

Leaves were assessed for black rot every 2–3 days by counting black rot lesions, and the average number of lesions per leaf was calculated. In total, 218 symptomatic leaves were assessed in 2013, 249 in 2014 and 511 in 2015. Because the final number of lesions per leaf differed among the 3 years, black rot severity on leaves (BRL) in each year was rescaled to the maximum number of lesions observed in that year. Thus, BRL was expressed on a 0–1 scale.

Clusters were assessed for black rot every 3–4 days. In total, 32 symptomatic clusters were assessed in 2013, 44 in 2014 and 61 in 2015. Each cluster was rated on a 0–1 scale[28] as follows: 0 (healthy clusters), 0.025 (≤5% of the cluster surface affected by black rot), 0.15 (5 to ≤25% of the cluster surface affected by black rot), 0.375 (25 to ≤50% of the cluster surface affected by black rot) and 0.75 (>50% of the cluster surface affected by black rot). Black rot severity on clusters (BRC) was then calculated by averaging the ratings of all clusters.

2.3 Model structure

The weather data were used to operate the model, as described in Rossi *et al.*[26] Briefly, the maturation of ascospores and conidia on overwintered berry mummies was calculated. A spore release event was predicted when Rain ≥ 0.5 mm in 1 h, and the corresponding inoculum dose (SPODOSE) was calculated by combining spore release, deposition and survival.[26] Following a spore release event, an infection period and infection severity (SEV′) were predicted based on temperature and wetness duration. Finally, the infection severity on leaves (SEVL) was calculated as SEVL = SPODOSE × SEV′. Beginning from the time when 80% of flowerhoods had fallen (BBCH 68), infection severity was also calculated for clusters (SEVC). Black rot lesion appearance was expected after an incubation window of 175–305 degree-days (calculated using 6 and 24 °C as cardinal temperatures); SEVL and SEVC were distributed over the incubation period by using the first derivative of the equation ONS (relative lesions onset) as described in Rossi *et al.*[26] Pycnidia were expected to appear on lesions in 262–392 degree-days (with 6 and 24 °C as cardinal temperatures), and the abundance of pycnidia (PYC), which served as sources of secondary inoculum, was calculated as a function of temperature.[26]

2.4 Data analysis

A Bayesian analysis was used to evaluate the correspondence between infection periods predicted by the model and black rot symptoms observed in the vineyard. Dates of disease assessments (or cases) were categorised as either 0 (disease did not increase relative to the previous assessment) or 1 (leaf lesion number or cluster severity increased relative to the previous assessment). Similarly, cases were categorised as either 0 or 1 based on whether or not the model predicted the appearance of new symptoms (new lesions or increased cluster severity). Cases were subsequently classified as: (i) true positive, when predicted infection = 1 and disease increase = 1; (ii) true negative, when predicted infection = 0 and disease increase = 0; (iii) false positive, when predicted infection = 1 and disease increase = 0; (iv) false negative, when predicted infection = 0 and disease increase = 1. A contingency table (2 × 2) was prepared in which the cells contained the true positive proportion (TPP or sensitivity), the true negative proportion (TNP or specificity), the false positive proportion (FPP) and the false negative proportion (FNP).[29]

The posterior probability of predicting disease occurrence based on model output was calculated as part of the Bayesian analysis.[29] The following posterior probabilities were considered: (i) the probability of disease increase when predicted by the model, $P(P+|O+)$; (ii) the probability of no disease increase when not predicted, $P(P-|O-)$; (iii) the probability of no disease increase when predicted (i.e. unjustified alarms), $P(P+|O-)$; (iv) the probability of disease increase when not predicted (i.e. missed real infections), $P(P-|O+)$. The prior probabilities of disease increase were calculated as the proportion of cases (assessment dates) with $P(O+)$ or without $P(O-)$ disease increase in the vineyard over the total cases. The overall accuracy of the predictions was finally calculated as the ratio between correct and total predictions. The analysis was performed separately for leaves and clusters.

A receiver operating characteristic (ROC) curve was used to investigate which thresholds used for predicting infection severity (either SEVL or SEVC) corresponded to observed infection. Because at any disease assessment date (case) the expected disease increase may have resulted from one or more predicted infections, SEVL (or SEVC) used as thresholds in the ROC analysis consisted of the sum of the infection severity values of all the infections that were expected to produce new symptoms on that date. Two curves were created (one for leaves and one for clusters) by plotting the true positive proportion (TPP or sensitivity) against the false positive proportion (FPP or 1-specificity) for each threshold. The closer the ROC curve is to the upper left corner of the plot, the greater the accuracy of the test.[30,31] The area under the ROC curve (AUROC) and its standard error were used to evaluate the analysis; AUROC

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values between 0.5 and 1 indicate that the variable (SEVL or SEVC in this case) is a good predictor of the response variable (actual disease increase in this case). The *P*-value was calculated as the probability that the AUROC is significantly different from 0.5 (i.e. the ROC curve coincides with the line of no discrimination, which has coordinates of 0,0 and 1,1). The statistical software SPSS (v.21.0; SPSS Inc., Chicago, IL) was used for the ROC analysis. A Bayesian analysis was conducted as described earlier for four thresholds: SEVL (or SEVC) of >0.001, >0.002, >0.005 and >0.01. These thresholds were chosen as representative of the values of SEVL and SEVC calculated for single infections events predicted in this work.

To validate the ability of the model to predict epidemic development through the season, predicted infection severities for leaves and clusters (SEVL and SEVC respectively) were accumulated between BBCH 11 and BBCH 85 for SEVL and between BBCH 68 and BBCH 85 for SEVC. Accumulated infection severities were then rescaled to the final value of the year, and named as ISL for leaves and ISC for clusters, on a scale of 0–1. ISL and ISC values were then regressed against the values of BRL and BRC, respectively, that had been observed in the vineyard. The null hypotheses that the intercept of the regression line equals zero and that slope equals 1 were tested using a *t*-test; when this test was not significant, both null hypotheses were accepted, and the model was considered a statistically accurate predictor of the actual observations.[32] The distribution of residuals of predicted versus observed values was examined to evaluate the goodness of fit; this distribution was assessed by the adjusted R^2, the root mean square error (RMSE), the coefficient of residual mass (CRM) and the concordance correlation coefficient (CCC).[33,34] RMSE represents the average distance of real data from the fitted line, and CRM is a measure of the tendency of the equation to overestimate or underestimate the observed values (a negative CRM indicates a tendency of the model toward overestimation).[34] The CCC was calculated as a measure of model accuracy;[35] CCC is the product of two terms: the Pearson product-moment correlation coefficient between observed and predicted values and the coefficient C_b (bias estimation factor), which indicates the difference between the best-fitting line and the perfect agreement line (CCC = 1 indicates perfect agreement).[34]

3 RESULTS

In 2013, the period of grapevine susceptibility to black rot began on 23 April (first leaf unfolded, BBCH 11) and ended on 30 August (softening of berries, BBCH 85); cluster susceptibility began on 8 June. During this period, 240.2 mm of rain fell mainly from late-April to late-May and in the second half of August. Between these two rainy periods, there were only four rainy days: 9 June (2.6 mm of rain), 27 June (13.6 mm), 14 July (1.00 mm) and 29 July (1.8 mm) (Fig. 1A). The model predicted a total of 19 infection periods (Fig. 1B). Most of these infection periods (13 periods) were predicted during the first rainy period (Fig. 1B) and only concerned leaves. The accumulated severity of these infections accounted for about 80% of the seasonal ISL (Fig. 1C). Disease symptoms caused by these predicted infections were expected to appear between 8 May and 15 June (Fig. 1B). Others symptoms were expected to appear in the last 10 days of June (because of the infection predicted on 10 June, which was about 10% of the final ISL) (Fig. 1C), in the second 10 days of July (because of the infection predicted on 28 June) and finally at the end of August (because of the few infections predicted after mid-August) (Fig. 1B). The latter two infection periods accounted for only a minor part of SEVL and SEVC (Fig. 1C). All season long, new black rot lesions appeared on leaves as predicted by the model (Fig. 1B). In substantial agreement with the model predictions, about 85% of the total lesions on leaves appeared within mid-June, and about 96% of them within early July (Fig. 1C). About 2% of leaf lesions appeared between late July and early August, and these were not predicted by the model (Figs 1B and C). In agreement with model predictions, black rot symptoms were first evident on clusters in early July, and the disease on clusters was minimal: the final disease severity index for clusters was 0.07 (Fig. 1C).

In 2014, the period of grapevine susceptibility to *G. bidwellii* infection began on 3 April (first leaf unfolded, BBCH 11) and ended on 1 August (softening of berries, BBCH 85); clusters were first susceptible on 30 May (80% of flowerhoods fallen, BBCH 68). A total of 121 mm of rain fell over the season, and the distribution was relatively regular (Fig. 2A). The model predicted 28 infection events (Fig. 2B). These predicted infections were also rather regularly distributed over the season, but two periods were notable: (i) the repeated infections predicted between 5 April (first seasonal infection predicted) and 4 May, with predicted onset of lesions on leaves between late April and late May; (ii) the predicted infection on 23 May, with symptoms predicted to appear on leaves between 5 and 12 June (Fig. 2B). These two periods accounted for 87% of the predicted accumulated infection severity on leaves (Fig. 2C). Field observation of disease on leaves coincided with the two periods predicted by the model (Fig. 2B), and the accumulated observed disease severity at the end of these periods was about 65% of the final value (Fig. 2C). The first cluster infection, which occurred on 31 May, and other cluster infections in the second half of June were correctly predicted by the model; these infections contributed to the increase in the predicted infection severity on both leaves and clusters (Figs 2B and C). The final disease severity index on clusters was 0.22.

In 2015, the first unfolded leaves (BBCH 11) were observed on 12 April, 80% of flowerhoods had fallen (BBCH 68) on 25 May and softening of berries (BBCH 85) was evident on 12 August. A total of 192.8 mm of rain fell over the growing season, mainly from early April until the end of May, around mid-June and in August (Fig. 3A). The model predicted a total of 16 infection events. Ten of these infections were predicted between 18 April and 27 May, and three were predicted to have high infection severity, with expected disease onset on leaves between early May and mid-June. Three additional infections were predicted between 15 and 29 June, and these were expected to produce symptoms on both leaves and clusters between late May and early July (Fig. 3B). The periods when disease symptoms were observed in the vineyard and the observed infection severity on both leaves and clusters agreed with model predictions (Figs 3B and C). Final disease was severe; BRL was 10 times higher in 2015 than in 2013 or 2014. BRC was 0.47; all clusters were affected by black rot, indicating a substantial loss of yield.

Bayesian and ROC analysis showed that the model was highly accurate in predicting the appearance of black rot symptoms on both leaves and clusters. The AUROC (Fig. 4) was 0.870 ± 0.029 for leaves and 0.770 ± 0.089 for clusters, and both were significantly greater than 0.5 ($P < 0.001$ for leaves and $P = 0.005$ for clusters). The use of SEVL or SEVC > 0 as a threshold for predicting an infection provided an overall accuracy of 0.804 for leaves and 0.769 for clusters (Table 1). The posterior probability of providing unjustified alarms, i.e. $P(P+|O-)$, was ≤0.180 for both leaves and clusters, and the probability of missing real infections, i.e. $P(P-|O+)$, was 0.175 for leaves and 0.263 for clusters. These unpredicted disease onsets

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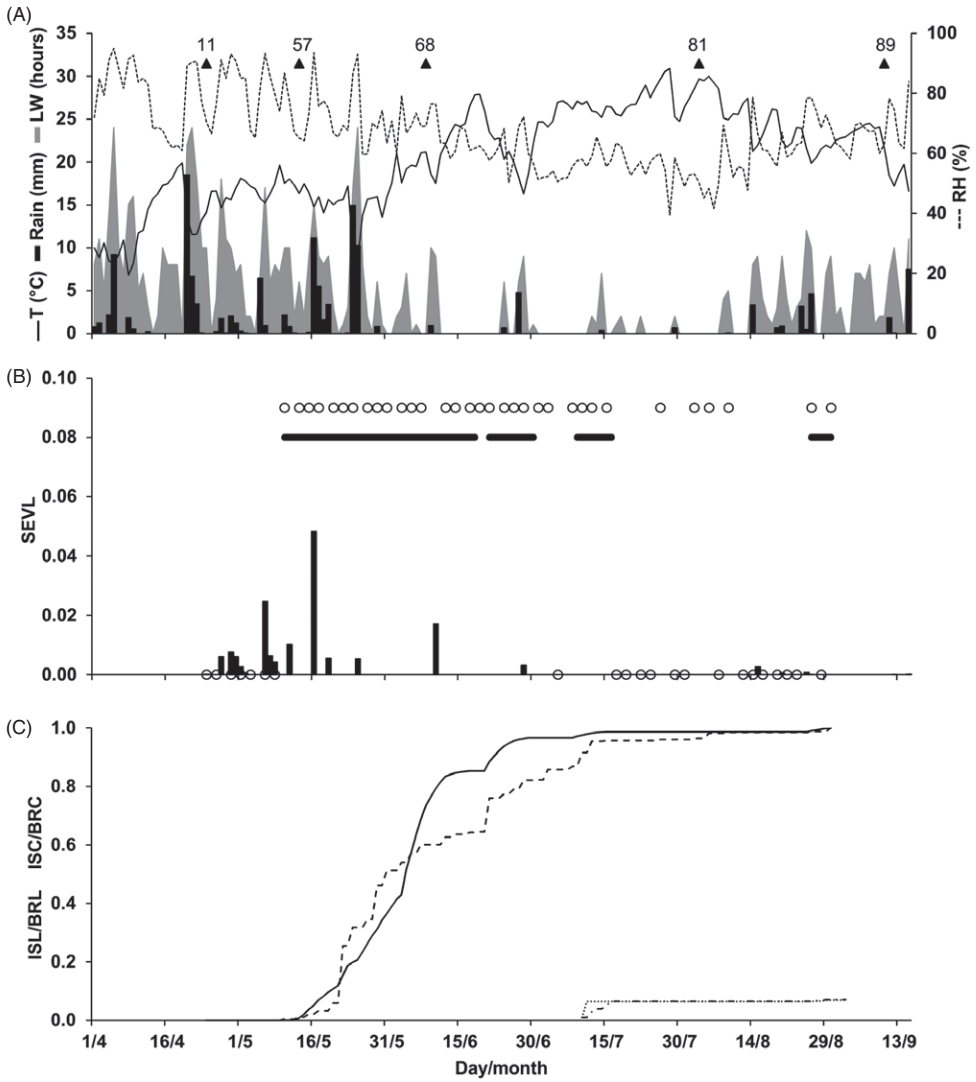


Figure 1. Model output for the campus vineyard at the University of Piacenza in 2013. (A) Weather variables: air temperature (*T*, °C) (solid line), rainfall (Rain, mm) (black bars), leaf wetness duration (LW, h) (grey area) and relative humidity (RH, %) (dashed line). Black triangles (▲) indicate the vine growth stages according to Lorenz *et al.*:[27] 11 = first leaf unfolded and spread away from shoot; 57 = inflorescences fully developed, flowers separating; 68 = 80% of flowerhoods fallen; 75 = berries pea sized and clusters hanging; 81 = beginning of ripening, and berries beginning to develop variety-specific colour; 89 = berries ripe for harvest. (B) Infection severity on leaves (SEVL) predicted by the model and as indicated by black bars. Thick black lines over the bars represent the expected appearance of black rot lesions after the incubation period. The empty circles (○) represent the results of the field leaf assessments: circles in the upper part of the graph indicate that new lesions were observed; circles on the bottom of the graph indicate that new lesions were not observed. (C) Accumulated infection severity predicted by the model and rescaled to the final value of the year (solid line for leaves and dash-dot line for clusters); accumulated value of black rot lesions on leaves (BRL) rescaled to the maximum number of lesions observed in that year (dashed line); black rot severity on clusters (BRC) calculated by averaging the ratings of all the clusters (dotted line). The maximum value of BRC was 0.071.

occurred in 18 cases (16 cases for leaves and two cases for clusters) and accounted for 1.3% of total leaf lesions and 8.3% of cluster severity. In 78% of these false-negative predictions (14 cases), the difference between predicted and observed disease onset was within ± 2 days; more specifically, it was ± 1 day in 11 cases and ± 2 days in three cases (Figs 1B, 2B and 3B). In four cases (22% of the total FNP), the difference was between 5 and 9 days; that occurred for the four disease assessment dates between 27 July 2013 and 9 August 2013, when 14 new lesions were observed on leaves but had not been predicted (Figs 1B and C). When SEVL or SEVC > 0.001 was used as a threshold, the overall accuracy of the model predictions increased, as did the probability of correctly predicting an increase in disease on both leaves and clusters. The probability of providing unjustified alarms, $P(P+|O-)$, also decreased, while the probability of missing infections, $P(P-|O+)$, did not decrease (Table 1). No improvements in model performance were observed by further increasing the infection severity threshold (Table 1).

In the linear regression of predicted versus observed data for black rot severity on leaves (Fig. 5A), the slope was not significantly different from 1 ($P = 0.186$), but the intercept was significantly different from zero ($P = 0.006$). On clusters (Fig. 5B), the slope was not significantly different from 1 ($P = 0.115$), but the intercept was not significantly different from zero ($P = 0.434$). For the goodness of fit of predicted versus observed data, $R^2 = 0.962$ and CCC = 0.974 for leaves, and $R^2 = 0.975$ and CCC = 0.982 for clusters. RMSE was low for both leaves and clusters (0.091 and 0.066 respectively). CRM indicated a slight overestimation of observed data for leaves (−0.077) and an underestimation for clusters (+0.060) (Fig. 5). For

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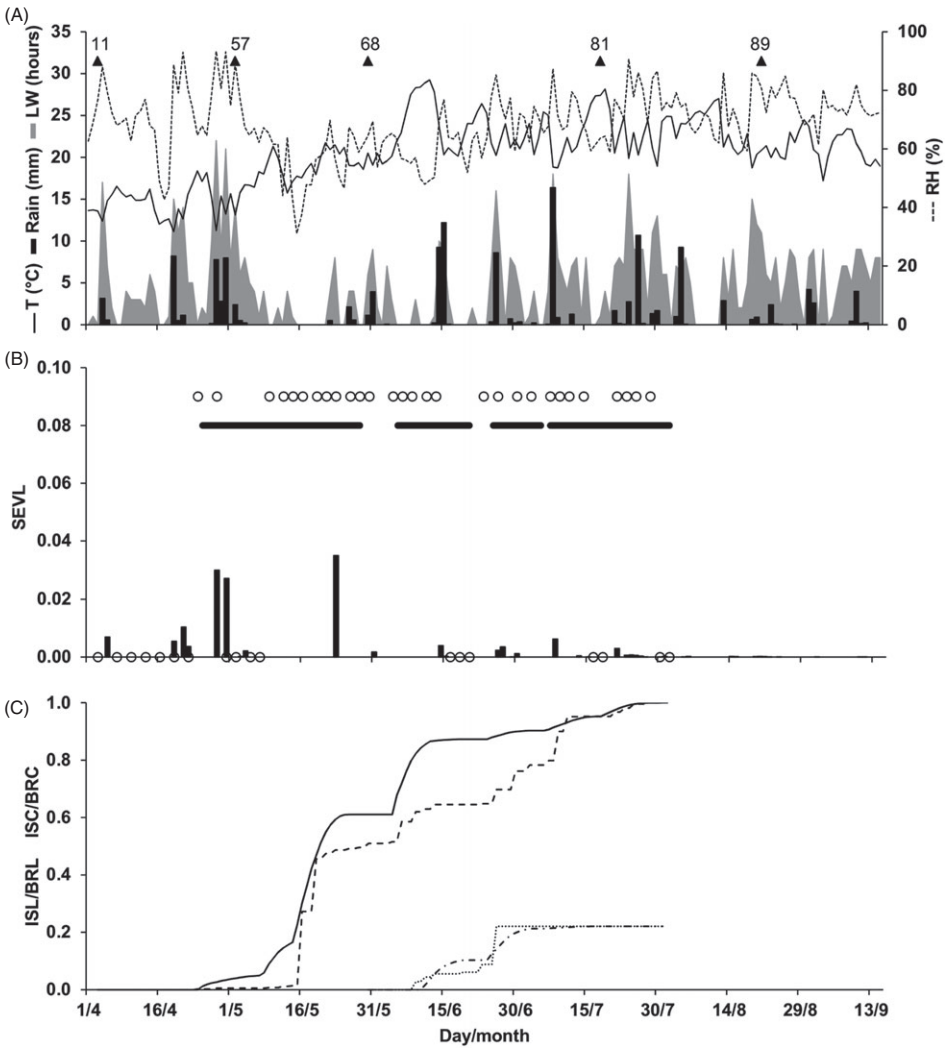


Figure 2. Model output for the vineyard on the campus of the University of Piacenza in 2014. (A) Weather variables: air temperature (*T*, °C) (solid line), rainfall (Rain, mm) (black bars), leaf wetness duration (LW, h) (grey area) and relative humidity (RH, %) (dashed line). Black triangles (▲) indicate the vine growth stages according to Lorenz *et al.*:[27] 11 = first leaf unfolded and spread away from shoot; 57 = inflorescences fully developed, flowers separating; 68 = 80% of flowerhoods fallen; 75 = berries pea sized, clusters hanging; 81 = beginning of ripening and berries beginning to develop variety-specific colour; 89 = berries ripe for harvest. (B) Infection severity on leaves (SEVL) predicted by the model and as indicated by black bars. Thick black lines over the bars represent the expected appearance of black rot lesions after the incubation period. The empty circles (○) represent the results of the field leaf assessments: circles in the upper part of the graph indicate that new lesions were observed; circles on the bottom of the graph indicate that new lesions were not observed. (C) Accumulated infection severity predicted by the model and rescaled to the final value of the year (solid line for leaves and dash-dot line for clusters); accumulated value of black rot lesions on leaves (BRL) rescaled to the maximum number of lesions observed in that year (dashed line); black rot severity on clusters (BRC) calculated by averaging the ratings of all the clusters (dotted line). The maximum value of BRC was 0.221.

leaves, overestimation mainly occurred when the observed ISL was >0.6 (Fig. 5A), which usually occurred between the end of May and early June (Figs 1C, 2C and 3C).

4 DISCUSSION

In this work, a recently published model[26] that predicts black rot epidemics was evaluated by comparing model output with observed data for disease on leaves and clusters during a 3 year period (2013 to 2015) in a vineyard in north Italy. The black rot epidemics observed in these 3 years differed in severity: numbers of leaf lesions were more than 10 times greater in 2015 than in 2013 or 2014, and the severity on clusters ranged from 7% in 2013 to 47% in 2015 (when all of the clusters suffered from black rot). This substantial range in disease levels enhanced the usefulness of the observed dataset for evaluating both model accuracy (i.e. the closeness of a predicted value to its 'true' value) and robustness (i.e. the capacity of the model to perform equally well across different epidemiological conditions).[36]

The model was evaluated by comparing predicted and observed disease data on both leaves and clusters. The model predictions were designated to be 'true positive' when symptoms were predicted to appear and really did appear as new lesions on leaves or increased disease severity on clusters. Similarly, the model was determined to have provided false negative or false positive predictions (which are both errors) based on this comparison. False negative predictions occur when the model does not predict a real infection and, consequently, a treatment should not be recommended when necessary. False positive predictions occur when an infection is predicted that really does not occur, resulting in an unnecessary treatment. Because the model predicts the appearance of disease on any day depending on (i) infection occurrence

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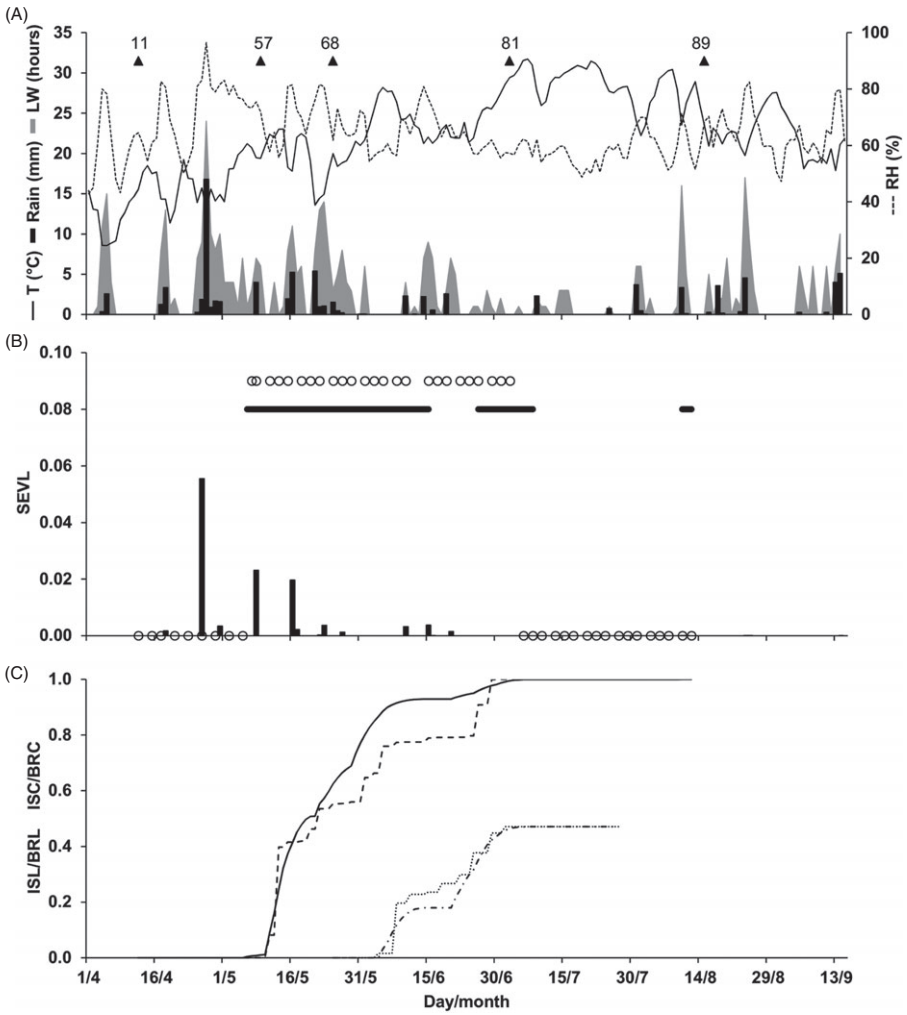


Figure 3. Model output for the vineyard on the campus of the University of Piacenza in 2015. (A) Weather variables: air temperature (*T*, °C) (solid line), rainfall (Rain, mm) (black bars), leaf wetness duration (LW, h) (grey area) and relative humidity (RH, %) (dashed line). Black triangles (▲) indicate the vine growth stages according to Lorenz *et al.*:[27] 11 = first leaf unfolded and spread away from shoot; 57 = inflorescences fully developed, flowers separating; 68 = 80% of flowerhoods fallen; 75 = berries pea sized, clusters hanging; 81 = beginning of ripening and berries beginning to develop variety-specific colour; 89 = berries ripe for harvest. (B) Infection severity on leaves (SEVL) predicted by the model and as indicated by black bars. Thick black lines over the bars represent the expected appearance of black rot lesions after the incubation period. The empty circles (○) represent the results of the field leaf assessments: circles in the upper part of the graph indicate that new lesions were observed; circles on the bottom of the graph indicate that new lesions were not observed. (C) Accumulated infection severity predicted by the model and rescaled to the final value of the year (solid line for leaves and dash-dot line for clusters); accumulated value of black rot lesions on leaves (BRL) rescaled to the maximum number of lesions observed in that year (dashed line); black rot severity on clusters (BRC) calculated by averaging the ratings of all the clusters (dotted line). The maximum value of BRC was 0.471.

and (ii) duration of incubation, model errors may have two main sources: (i) errors in predicting infection occurrence and (ii) errors in predicting incubation duration.

Overall, the model accurately predicted disease onset. The probability of predicting a disease increase that did not occur, i.e. $P(P+|O-)$, was ≤0.180 for both leaves and clusters, while the probability of failing to predict a disease increase that really did occur, i.e. $P(P-|O+)$, was 0.175 for leaves and 0.263 for clusters. This meant that the model failed to predict only a few of the increases in disease in the vineyard. Because the difference between expected and actual disease onset in most of these false negative predictions was ±2 days, most of these errors were probably caused by inaccuracies in calculating the incubation duration rather than errors in predicting infection occurrence. Assuming that a difference of ±2 days between predicted and actual disease onset was caused by errors in the incubation duration, the only infection not predicted by the model was the infection that caused the appearance of few new leaf lesions (14 lesions) between late July and early August 2013. Model accuracy was increased by using a threshold of >0.001 for predicting an infection period; more specifically, the probability of predicting an infection that did not occur was lower when the threshold was >0.001 rather than >0.

In the model used in this research, an incubation period ended when 175 degree-days (using 6 and 24 °C as the cardinal temperatures) were accumulated after infection,[37] and symptoms continued to appear until 305 degree-days were accumulated.[14] Molitor *et al.*[37] found that disease symptoms first became visible on leaves when 165.6–198.2 degree-days had accumulated after infection. A similar incubation duration was observed for clusters when clusters were inoculated between growth stage BBCH 68 (80% of flowerhoods fallen) and BBCH 77 (berries beginning to touch); the incubation duration was longer for later inoculation dates in Riesling vines[37] because of ontogenic or age-related resistance.[14] This

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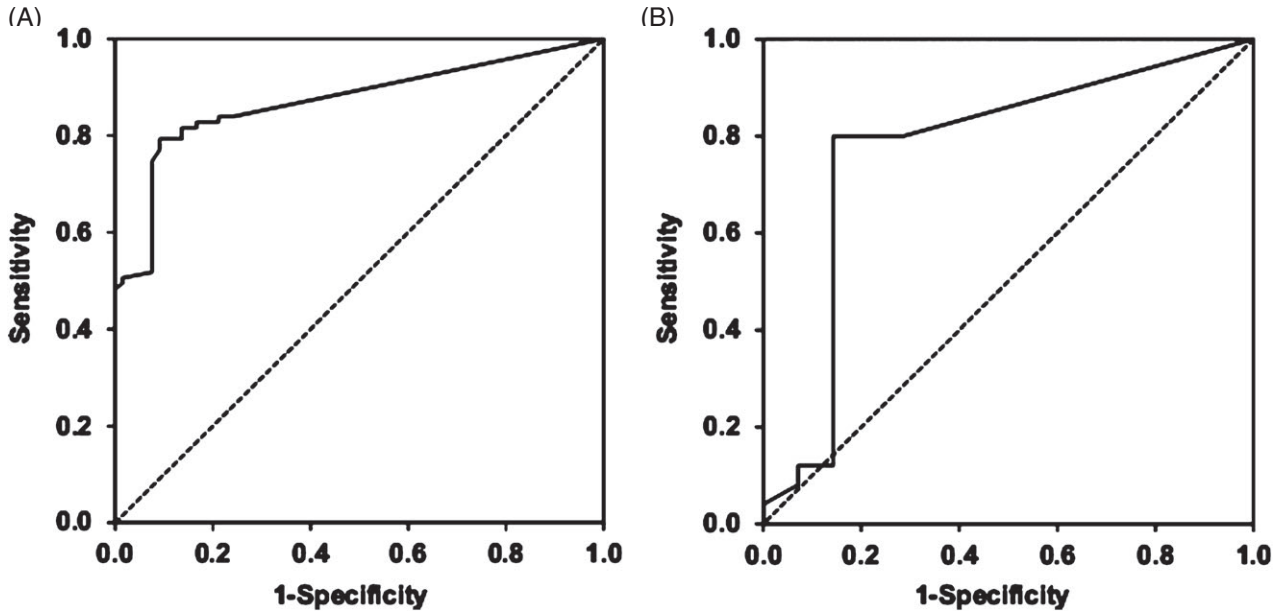


Figure 4. Receiver operating characteristic (ROC) curve for predicting black rot occurrence (i.e. disease increase) based on the model output for new leaf lesions (A) or increased cluster severity (B) during 2013, 2014 and 2015 at the campus vineyard of the University of Piacenza. The dotted line (0,0; 1,1) represents no discrimination.

Table 1. Evaluation of the infection severity calculated by the model as a predictor of the appearance of *G. bidwellii* symptoms on leaves and clusters. The predicted and observed data are based on the monitoring of a vineyard in Piacenza (north Italy). The vineyard was monitored between the first-leaf-unfolded stage and veraison in 2013, 2014 and 2015

	Proportions[b]					Posterior probabilities[d]			
Predicted infection severity[a]	TPP	FNP	FPP	TNP	Overall accuracy[c]	(P + |O+)	(P − |O−)	(P + |O−)	(P − |O+)
Leaves									
>0	0.839	0.161	0.242	0.758	0.804	0.820	0.825	0.180	0.175
>0.001	0.828	0.172	0.182	0.818	0.824	0.857	0.815	0.143	0.185
>0.002	0.793	0.207	0.091	0.909	0.843	0.920	0.786	0.080	0.214
>0.005	0.644	0.356	0.076	0.924	0.765	0.918	0.680	0.082	0.320
>0.01	0.471	0.529	0.000	1.000	0.699	1.000	0.589	0.000	0.411
Clusters									
>0	0.800	0.200	0.286	0.714	0.769	0.833	0.737	0.167	0.263
>0.001	0.800	0.200	0.143	0.857	0.821	0.909	0.737	0.091	0.263
>0.002	0.600	0.400	0.143	0.857	0.692	0.882	0.583	0.118	0.417
>0.005	0.320	0.680	0.143	0.857	0.513	0.800	0.452	0.200	0.548
>0.01	0.080	0.920	0.071	0.929	0.385	0.667	0.378	0.333	0.622

[a] Thresholds of the infection severity predicted by the model used as cut-off values.
[b] TPP (true positive proportion, or sensitivity): symptoms were observed when predicted by the model; FNP (false negative proportion): symptoms were observed but were not predicted by the model; FPP (false positive proportion): symptoms were not observed but were predicted by the model; TNP (true negative proportion, or specificity): symptoms were not observed and were not predicted by the model.
[c] Overall accuracy calculated as the proportion of correct predictions.
[d] Posterior probabilities: $P(P+|O+)$ = disease increased when increase was predicted; (ii) $P(P-|O-)$ disease did not increase when increase was not predicted; $P(P+|O-)$ = disease did not increase when increase was predicted; $P(P-|O+)$ = disease increased when increase was not predicted. Prior probabilities for disease increase (O+) and no increase (O−) were $P(O+) = 0.569$ and $P(O-) = 0.431$ for leaves, and $P(O+) = 0.641$ and $P(O-) = 0.359$ for clusters.

variability in incubation duration strongly supports our inference that most of the false negative predictions of infection in the present work were related to variability in incubation duration rather than to errors in predicting infection occurrence. Even though the prediction of the incubation period can be improved, associated with the diversity of grapevine varieties, this error in predicting the incubation period may not reduce the ability of the model correctly to predict infection periods or reduce the value of the model for the timing of fungicide applications.

When used to predict the development of black rot epidemics, the model slightly overestimated disease severity, mainly on leaves. This overestimation largely occurred in the second part of the epidemic, when the observed disease severity was >0.6. This result was likely caused by an overestimation of the predicted

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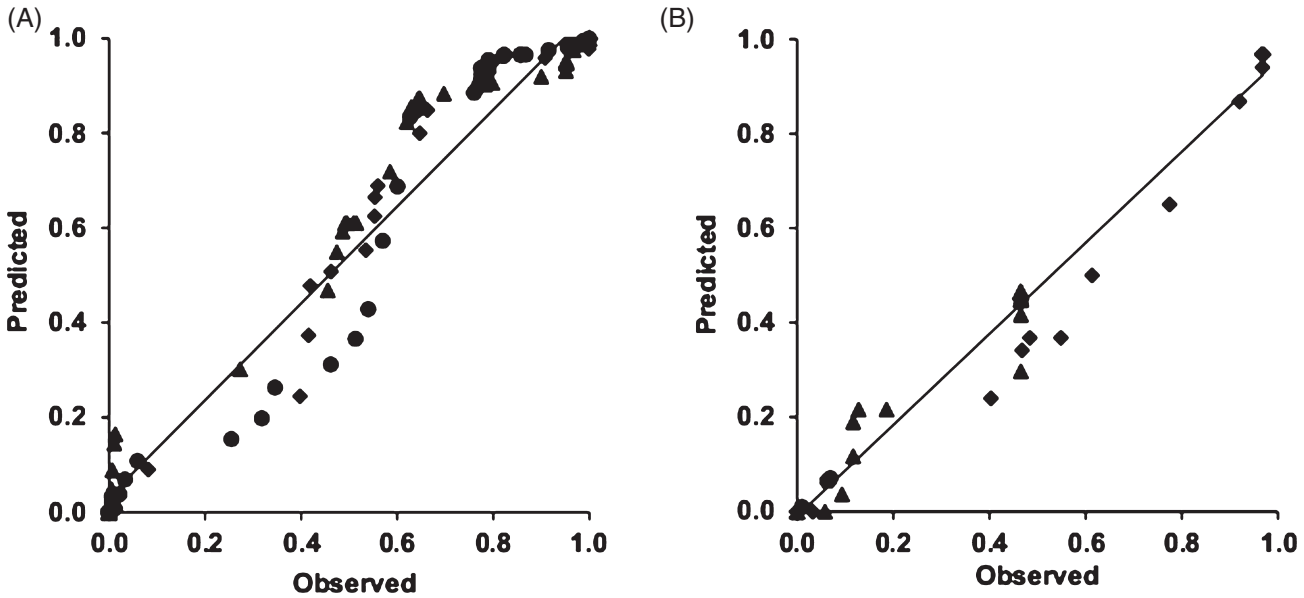


Figure 5. Plots of predicted versus observed values of black rot infection severity for leaves (A) and clusters (B) in 2013 (●), 2014 (▲) and 2015 (♦) at the campus vineyard of the University of Piacenza. Predicted values are accumulated values of infection severity for leaves and clusters (ISL and ISC respectively) through the growing season and are rescaled to the final value of the year. Observed values in (A) are the accumulated number of lesions on leaves (BRL) rescaled to the maximum number of lesions observed in that year. Observed values in (B) are the accumulated values for infection severity (BRC) on clusters rescaled to the final value of the year.

infection severity in the first part of the epidemic, i.e. in April and May. The model calculates infection severity by combining, for any infection period, the available inoculum dose and the infection efficiency of this inoculum. The inoculum dose consists of both primary and secondary inoculum. Availability of primary inoculum is determined by equations developed on the basis of field data collected in vineyards in Michigan,[13] New York[19,38] and Switzerland.[5] Availability of secondary inoculum is mainly based on assumptions about the production of pycnidia and conidia on lesions, because no precise data are available in the literature for *G. bidwellii*. Infection efficiency is calculated by an equation developed on the basis of the data obtained by Molitor[39] in environmentally controlled experiments. In those experiments, artificially inoculated leaves were kept at constant temperatures between 10 and 30 °C, with 4–24 h of wetness duration, and clusters were kept at fluctuating temperatures between 17.3 and 25.1 °C, with 6–72 h of wetness duration. Perhaps the discrepancies between predicted and actual infection severities in the present work were caused by differences in inoculum availability in the considered vineyard versus inoculum availability estimated in the model, rather than by differences in the response of infection efficiency to temperature and wetness duration. Further studies on the dynamics of both primary and secondary inoculum should therefore improve the accuracy of the model in predicting infection severity.

Different methods have been developed for improving the scheduling of fungicide applications against black rot.[17,22–25] Most of these methods have been based on the work of Spotts,[24] who developed a table describing the combinations of hours of wetness and temperatures necessary for *G. bidwellii* infection. The author tested the use of this table for two seasons in a vineyard in Ohio, which resulted in the prediction of a high number of infection periods (54 and 34 infection periods in the two seasons) and the need for a high number of fungicide treatments (8 and 12).[24] By using the relationships provided by Spotts[24] to predict infection periods and to schedule fungicide treatments, Ellis *et al.*,[17] Maurin *et al.*[22] and Smith[40] obtained good control of the disease and a reduction in fungicide usage compared with calendar applications. Gadoury *et al.*[41] also developed a set of simple rules to define the criteria for black rot infection based on a combination of wetness duration and temperature. Recently, Molitor *et al.*[23] developed a disease prediction system based on suitable conditions for infections[39] and duration of incubation.[37] The model was validated against 330 observed disease assessment data on leaves only; the model failed to predict a true infection in 41 of the 165 assessments with new disease symptoms and predicted an infection that did not occur in 79 of the 165 assessments with no new disease symptoms.

The model developed by Rossi *et al.*[26] and validated in the present work represents an improvement of the previous models because it considers not only infection periods but also the production, dispersal and survival of the inoculum. In addition, the model quantifies infection risk, while the Spotts-based methods only predict whether an infection can or cannot occur. The model of Molitor *et al.*[23] distinguishes light, moderate and severe infections, but that aspect of the model has not been validated. The present work demonstrates that the model of Rossi *et al.*[26] is highly accurate and robust in predicting the infection periods and dynamics of black rot epidemics. It follows that the model can be used for scheduling fungicide sprays in vineyards. If the model output had been used to schedule fungicide applications in the vineyard of the present study, fungicides would have been applied 5 times in 2013 and 2014 and 4 times in 2015; calendar-based timing, in contrast, would have resulted in nine, ten and eight applications in these years. The model of Rossi *et al.*[26] could also be integrated into vite.net, which is a decision support system that already includes models for downy and powdery mildews and Botrytis bunch rot.[42,43] The future addition of the black rot model to vite.net should make it possible to schedule fungicide applications and to select the most appropriate fungicides by simultaneously considering the risk of the four main diseases of vineyards.

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